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What is ESGF Or ESGF2-US?

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What is the Earth System Grid Federation?

The Earth System Grid Federation (ESGF) is a global peer-to-peer network of data servers that archives and distributes the planet's largest collection of Earth system model output for thousands of climate and environmental scientists worldwide. After recently providing data from a variety of climate change scenario projections to inform the United Nations Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6), ESGF will be re-architected to improve its reliability, scalability, and usability for new generations of high resolution global Earth system model outputs from dozens of international modeling centers.

What is ESGF2-US?

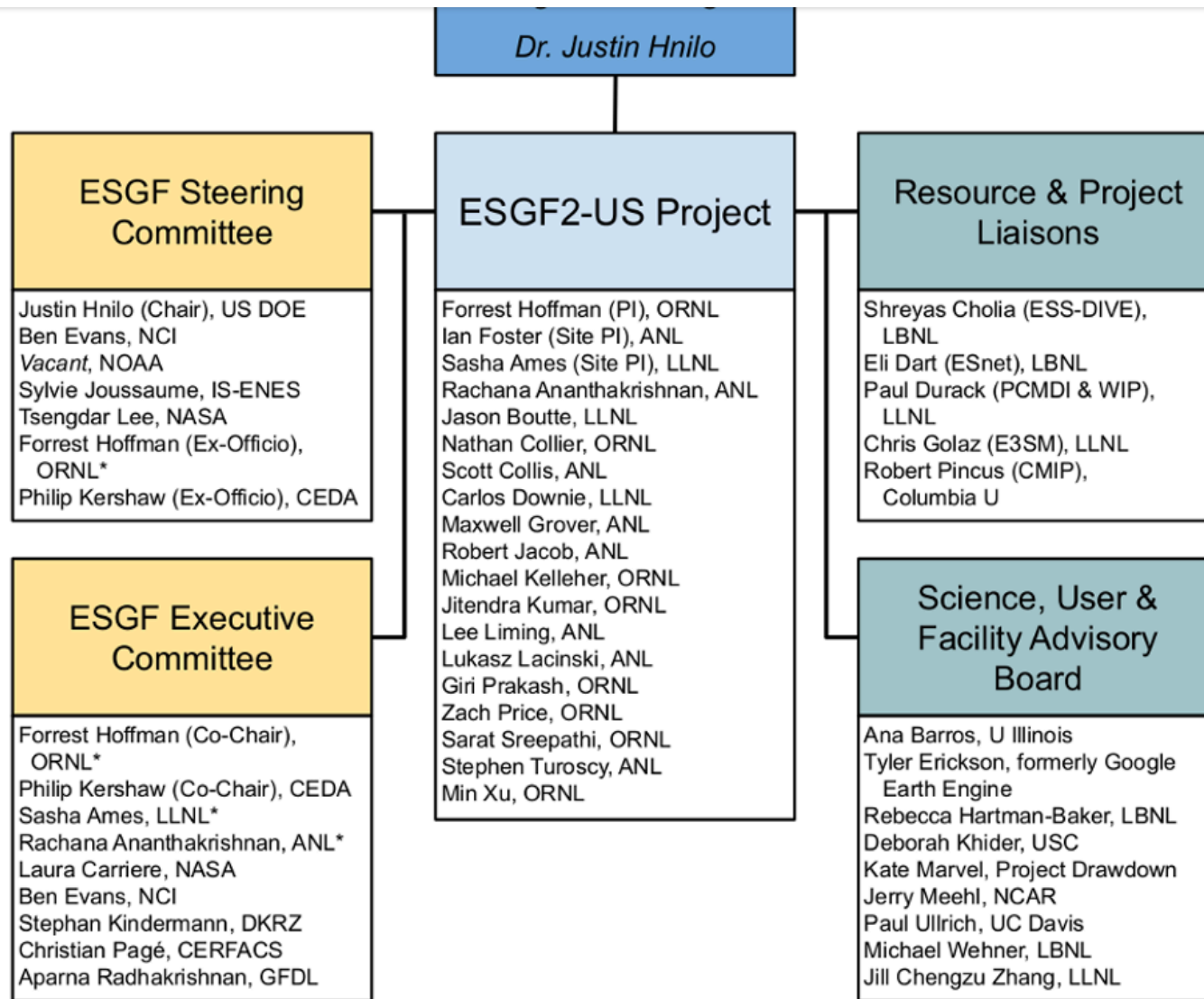
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spanning, international collaboration.

The ESGF2 Project—which brings together Earth, environmental, and computer scientists from Oak Ridge National Laboratory, Argonne National Laboratory, and Lawrence Livermore National Laboratory—will deploy infrastructure to move data as needed among US and international high performance computing centers and across simulation and analysis platforms to enhance the productivity of Earth system scientists. In addition to building out a new data backplane infrastructure, we will employ cloud technologies for offering integrated compute and storage resources to enable “compute-near-the-data” analysis capabilities for climate research supported by the US Department of Energy and the broader research community. In this presentation, we will describe plans for development and deployment of the next generation ESGF and highlight opportunities for community engagement in the design and operation of this international Earth science resource.

How is ESGF2-US Structured?

The ESGF2-US adopted the following governance structure:


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*Indicates ESGF2-US team member

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